

8 2026

dobachi

2026-08-12

11.4 (b) — 25

11.5 (c) — 25

11.6 25

11.7 26

- : 2026-08-01
- : 2026 7 28 16:27 M7.1 7 8
-

grounded-research

check_spans.py

- : > - [S-xx]
- :
- : §8

0.

2026 7 28 7 8 [S-01] 7 2016 10 3 3 [S-01]

36 179 8 1 11 30 [S-01] 465 9,872 [S-52]

§1 §2

3

3

TSMC 7 [S-01] 5 [S-01] → §1.1

§3.1

[S-24] → §3.2 §3.3

[S-52] → §5

§10

§8

:

	§1
	§2
	§3
	§4
	§5
SNS	§6
IT	§11
	§8 §10

1.

Mj7.1 16km 7

Mj 7.1 16 km 7 — [S-01]

7 28 19 30 8 — [S-01]

10 3 3 7

28 2016 4 14 16 10 3 7 3 7 — [S-01]

2,438 Gal

2,438 Gal KiK-net 1,667 Gal — [S-01]

1.1 —

fab 5 6

6 — [S-01]
5 — [S-01]

7	—
6	
6	
6	
5	JASM(TSMC)

5

TSMC 5 1 — [S-42]

fab fab

1.2

— [S-01]
3 163

163 29 8 00 — [S-02]

2.

2.1 49 2026 8 11 16 00

49 8 11 16 00 8 1 11 30 [S-56]

 — [S-56] — [S-56]
8,611 330 8,603 327 [S-56]

39 26 327 353 392 1,133 1,045 8,603 10,781 — [S-56]

8/1 11:30		
39	39	36
26	27	6
327	330	90
1,133	1,133	179
1,045	1,045	240
8,603	8,611	1,021

—

47 8 9 10 00 [S-57]

— [S-57] 39 23 146 169 208 699 1,045 7,622 9,366 — [S-57]

8/1 11:30		36	6	90	179	240	1,021
8/9 10:00 47	39	23	146	699	1,045		7,622
8/11 16:00 49	39	26	327	1,133	1,045		8,603

- 8 9 8 11 39
- 2 1,045
- 8 11 8/9→8/11 2 62% 2.2

2 49 39 2 2016

10 + + 1,440 → 10,781 7.5

2

§9

:

36 6 90 179 240 1,021 — [S-01] 36 6 93 179 240 1,029 — [S-01]

1 8 1 [S-56]

49 :

— [S-56] — [S-56] — [S-56]

36 1 2 39 [S-56] 49 3

8 1 34 36 2 :

20 2 7 1 4 36 — [S-01]

2 49

— [S-56] — [S-56]

— [S-56]

2.2

:

12 7 — [S-01]

:

31 11 9 — [S-01]

:

28 3 — [S-01]

:

29 7 8 — [S-01]

2.3

84,000 **5 23**

84,000 — [S-03] 5 7/28 283 439 — [S-03]

4.9 PDF

4.9 31 () — [S-04] www.meti.go.jp HTTP 403 CAPTCHA

Around 48,300 households in Kumamoto Prefecture lost power — [S-05]

2.4

29 12 465 9872 — [S-01]

11 9

17 30 8 — [S-01] 42 8 8 3,600 — [S-01]
11 9 — [S-01]

3. — fab

3.1

JASM TSMC

— [S-11] 2 29 — [S-11]

— 8 4 **2026-08-12** **2 7 31** 8 1 [S-59]

8 4 8 — [S-06]

— **8 4** 4 [S-58]

8 5 — [S-58] 3 — [S-58]

§3.4 3 **8 12**

3 8 5

7 29 — [S-07] 8 5 3
— [S-07]

— [S-10]

JASM [S-11]

— [S-08] — [S-09]

— [S-12]

3.2 —

10 — [S-21] — [S-21]
7 31

3 8 5 — [S-16] 31 8 3 7 — [S-16] 31 2026
[S-18] 8 5 2 8 7 — [S-19] 32 7 —
— [S-20] — [S-18]

3.3 —

1 — [S-13] — [S-13]
— [S-14]

3 110.8km

110.8km — [S-54] E3 ()IC IC — [S-54] E3A
()JCT ()IC — [S-54] E77 JCT TB — [S-54]

3.4

but the impact on the global chip supply chain is expected to be limited, according to forecasts. — [S-23] TSMC's Kumamoto fab accounts for less than 3% of the company's total production capacity — [S-24] Analysts say the chip plants suffered reduced damage because seismic-resistant designs were put in place in advance, given Japan's frequent earthquakes. — [S-23]

— [S-26]

— [S-45] Even if wafer fabs avoid major damage, disruptions to transportation, power supply, or logistics could still result in short-term supply chain delays. — [S-24]

fab

fab

§3.2 3.3

4. fab

4.1

JASM 2021 11 9 **2016** **5 7**

20 70 8,000
— [S-32]

— [S-32] 2021 11 9 SSS 5 570

:

100 TSMC — [S-34]
PDF 403 :

— [S-31]

4.2 —

JASM §8

— [S-35] — [S-36] 30

— [S-38]

— [S-36]

— [S-37]

4.3

V-Fine 6 100 200gal — [S-39]
— [S-39]

— [S-40] — [S-40] 0.6 600gal

TSMC :

The clean rooms are isolated from external vibrations with double-shell structures, and there are viscous dampers and hydraulic pistons installed aplenty. — [S-41]

2,438 Gal 6 100 200gal

4.4

2011

— [S-48] 1 2,500 — [S-49]

$$6 \rightarrow 6 \quad \text{--- [S-49]} \quad 0.5 \quad \text{--- [S-49]}$$

2016 → **2026**

2016 3 — [S-50]

8 [S-06] §3.1 **3** → **2**

16 — [S-44]

TSMC

Taiwan experienced an earthquake on April 3 reaching magnitude level 7.2 (the largest in Taiwan over the past 25 years). The maximum magnitude measured at the Hsinchu, Longtan, and Zhunan science parks was 5, and reached magnitude 4 at the Taichung and Tainan science parks. — [S-51] overall tool recovery of our fabs reached more than 70% within 10 hours of the April 3 earthquake, with new fabs such as the Fab 18 facility reaching more than 80%. — [S-51]

5 5

4.5 —

7 28 1 — [S-44]
— [S-44]
2 [S-11] §3.1

— [S-22]

5.

28 2 64 — [S-52] minori 58 —
[S-52]

465 9,872

6. SNS

8 SNS — [S-53]
7 29 10 30 7 30 [S-53]

AI 2024

7. Source Register

Tier	T1=	T2=	T3=	
ID	Source		URL	Tier
S-01	(2026) - Wikipedia		https:// ja.wikipedia.org/ wiki/ _%282026 %29	T3
S-02	7/29 08	3	https:// www.pref.kumamoto.jp/ uploaded/ attachment/ 315415.pdf	T1
S-03	7/29 6:00		https:// www.mlit.go.jp/ common/ 002014126.pdf	T1
S-04	8 15:30	8/1	https:// www.meti.go.jp/ 2026_ku- mamoto/pdf/ 20260801_1.pdf	T1
S-05	2026 Kumamoto earthquake - Wikipedia (English)		https:// en.wikipedia.org/ wiki/2026_Ku- mamoto_earthquake	T3
S-06	7/31	2	https:// www.sony- semicon.com/ja/ info/ 2026/2026073101.html	T2
S-07		3	https:// www.renesas.com/ ja/about/ newsroom/ impact-26- earthquake- kumamoto-3	T2

ID	Source	URL	Tier
S-08	7/28	https:// www.tel.co.jp/ news/disaster/ 2026/20260728_001.html	T2
S-09	2 7/29	https:// www.tel.co.jp/ news/disaster/ 2026/20260729_001.html	T2
S-10	EE Times Japan	https://ee- times.itmedia.co.jp/ ee/articles/ 2607/28/ news101.html	T3
S-11	ITmedia NEWS TSMC	https:// www.itmedia.co.jp/ news/article/ 2607/30/2000000305/	T3
S-12		https:// kumanichi.com/ articles/2019900	T3
S-13		https:// kumanichi.com/ articles/2020508	T3
S-14		https:// www.nikkei.com/ article/ DGXZQOUC30AXO0Q6A730C2000000/	T3
S-16	3 7	https:// www.nikkei.com/ article/ DGXZQOFD311MZ0R30C26A7000000/	T3
S-18		https:// www.nikkei.com/ article/ DGXZQOUC312930R30C26A7000000/	T3
S-19	8 7	https:// kumanichi.com/ articles/2020503	T3

ID	Source	URL	Tier
S-20		8 5 https://carview.yahoo.co.jp/news/detail/3144bc08f34ad666d4d5f68b9ead5b79c4b64b5d/	T3
S-21		https://www.nikkei.com/article/DGXZQOFD29B3M0Z20C26A7000000/	T3
S-22		https://chemi-caldaily.com/archives/835514	T3
S-23	Seoul Economic Daily Impact on Chip Supply Chain Seen Limited	https://en.sedaily.com/international/2026/07/29/kumamoto-quakes-impact-on-chip-supply-chain-seen-limited	T3
S-24	TrendForce 7.1 Kumamoto Earthquake: TSMC Confirms JASM Safe	https://www.trendforce.com/news/2026/07/29/news-7-1-kumamoto-earthquake-tsmc-confirms-jasm-safe-tel-halts-plants-as-chip-supply-chain-assesses-impact/	T3
S-26		https://www.nikkei.com/article/DGXZQOUC293G30Z20C26A7000000/	T3

ID	Source	URL	Tier	
S-31	JASM 2	https:// www.meti.go.jp/ policy/ mono_info_service/ joho/laws/ semiconductor/ semiconduc- tor_plan/ nin- tei_tokuteihandoutai_keikaku06.pdf	T1	403
S-32	2021 11 9	https:// www.sony- semicon.com/ja/ news/ 2021/2021110901.html	T2	
S-34		https:// www.reri.or.jp/ report-and- news/ ckm-kumamoto- reasonwhy/	T3	
S-35	2022 11	https:// www.cas.go.jp/ jp/seisaku/ keizai_anzen_hosyohousei/ r4_dai4/ siryou1.pdf	T1	
S-36	No221 2023.1	https:// www.sangiin.go.jp/ japanese/annai/ chousa/ keizai_prism/ backnumber/ r05pdf/ 202322101.pdf	T1	
S-37	6 2 9	https:// www.mext.go.jp/ content/ 20240209-mxt- jyohoka01- 000033848_01.pdf	T1	

ID	Source	URL	Tier
S-38		5 6 https://www8.cao.go.jp/cstp/gaiyo/yusikisha/20230622/siry2.pdf	T1
S-39		https://www.kajima.co.jp/tech/e_device/risk/tasoukai/index.html	T2
S-40		https://www.hirakata-g.co.jp/product/tackgel/	T2
S-41	Tom's Hardware TSMC factories rocked by magnitude 7.0 earthquake	https://www.tomshardware.com/tech-industry/semiconductors/tsmc-chipmaking-factories-rocked-by-magnitude-7-0-earthquake-that-was-the-strongest-in-27-years-but-facilities-escaped-unharmed-companys-earthquake-protection-measures-pay-off	T3
S-42	+IT	https://www.sbbbit.jp/article/st/186349	T3

ID	Source	URL	Tier
S-44		16 https://news.yahoo.co.jp/articles/0d8c5473327ed649ff2dd27af98254c10db46b2f	T3
S-45		https://news.yahoo.co.jp/articles/d16da3f04e4e7cadb6dbb6e880cbe625826bcb19	T3
S-48	EE Times Japan	https://ee-times.itmedia.co.jp/ee/articles/1604/18/news117.html	T3
S-49		http://www.cocn.jp/material/ec9a4f2565e27e752d082827591d868e213ba413.pdf	T2
S-50	ITmedia	8 https://www.itmedia.co.jp/news/article/2607/31/2000000341/	T3
S-51	TSMC Updated Statement on April 3rd Earthquake SEC Form 6-K	https://www.sec.gov/Archives/edgar/data/1046179/000104617924000038/a20240405.htm	T1
S-52		https://kumanichi.com/articles/2019886	T3
S-53	piyolog SNS	8 https://piyolog.hatenadiary.jp/entry/2026/07/30/143946	T3
S-54	NEXCO 4 7/29 11:25	https://www.w-nexco.co.jp/emc/20260729112506.html	T1

ID	Source	URL	Tier
S-55	Open Data Spaces RAM IPA DADC / ODS-RAM V2	https://open- dataspaces.gitbook.io/ ods-docs/jp/ ods-ram/ glossary.md	T1
S-56	49 R8.8.11 16:00	https:// www.fdma.go.jp/ disaster/info/ items/ 20260728ku- mamoto- jishin49.pdf	T1
S-57	47 R8.8.9 10:00	https:// www.fdma.go.jp/ disaster/info/ items/ 7595b20c947c985d45f6a34374b5f9ef31efbf94.pdf	T1
S-58	8 4 8/4	https:// www.renesas.com/ ja/about/ newsroom/ impact-26- earthquake- kumamoto-4	T2
S-59	2026-08-12 2	https:// www.sony- semicon.com/ja/ info/	T2

8.

JASM	TSMC	8	NOT-FOUND
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TSMC	NOT-FOUND NOT-FOUND NOT-FOUND 2
	NOT-FOUND

:

9.

- — 34 — 36 → 2026-08-12 49 1 2 §2.1
- 4.9 403
- PDF S-31 403
- 7/28
- 49 103,922 219,027 PDF
- 2026-08-12 47 49 4 2 7/31 [S-59]
- : 8/4 JASM TSMC §3.3 8/1
- 2 §2.1
-
-
- §3.4

10.

2026-08-01 08-02 S-54 08-12 S-55 S-56

	107
FOUND	105
NOT-FOUND	0
FETCH-FAIL	2 S-04 S-31 — 403

※ S-59 2
2026-08-12 17 S-55 4 S-56 9 S-57 2 S-58 2 S-56 S-57 PDF
pdftotext -layout

2026-08-12

citation_presence 0 → 12 12
12 §X ...
12 §0 §9 §10 §11
grounded-research :

```
python3 ../grounded-research/scripts/check_spans.py report.md
```

2026-08-02 : check_spans.py UTF-8 Shift_JIS
NOT-FOUND S-10 EE Times Japan S-10
NOT-FOUND
3 NOT-FOUND ... [134]

- span HTML: NFKC PDF: pdftotext
- : EE Times Shift_JIS UTF-8 NOT-FOUND ,
- 6 SEC EDGAR User-Agent www.meti.go.jp pr.tsmc.com 403
- : SNS

11.

3
IDS / Gaia-X / Catena-X

11.1 —

3

#				
(a)		§3	fab	§3.2
			§3.3	—
(b)		§2.1		34 36
			49	
(c)		§6	AI	

3

(a)

(b)

(c)

11.2

ODS-RAM Open Data Spaces Reference Architecture Model IPA DADC

Data Free Flow with Trust
— [S-55]

Open Data Spaces
— [S-55]

ODS Protocols ODP

Open Dataspaces

11.3 (a)
—

§3.2
§3.4

11.4 (b) —

§2.1 8 1 34 36
2026-08-12 49 1 2 §2.1

ODS-RAM

unified meta identifier UMI — [S-55]

2
ODS-RAM

Identity Layer: L3 — [S-55]

11.5 (c) —

§6 AI

> - [S-xx] §10

11.6

- 3
- 3
- (b)
-

11.7

- ODS-RAM 2026-08-12

(c) ODS-RAM

- §11